

CodeSquad Portfolio

About Us

At CodeSquad we specialize in Automation, SaaS, and AI-powered systems that help businesses work smarter, scale faster, and achieve more. From workflow automation and lead management to intelligent SaaS platforms and AI-driven solutions, we transform complex processes into seamless, efficient experiences. Our solutions span industries including healthcare, finance, education, immigration, e-commerce, and marketing, delivering tools that boost productivity, enhance decision-making, and accelerate growth. We don't just build software we build intelligent systems that empower your business to thrive in the digital era.

Our Services

- **Automation & Workflow Systems:** Streamline repetitive tasks, optimize operations, and manage leads with smart automation. Projects: Glick Automation, Job Nimbus, Job Nimbus Phase 2
 - **SaaS Platforms:** Deliver scalable, cloud-based solutions that improve collaboration, management, and user experiences. Projects: Mamas Compass, Meade Rewards, Settle In
 - **AI Systems & Intelligent Solutions:** Leverage AI for personalized experiences, analytics, language learning, healthcare monitoring, and smart decision-making. Projects: Gaeilgeoir, Frontline Dental, RTM, DropPR
 - **Marketing & Outreach Automation:** Automate your campaigns, social media posting, and lead generation to reach the right audience at the right time. Projects: Cold Email Outreach, Social Media Post Automation.
 - **E-commerce & Affiliate Optimization:** Aggregate products, compare intelligently, and optimize affiliate strategies for better reach and revenue. Projects: Mamas Compass, DropPR.
 - **Web Development:** Scalable and high-performance digital experiences.
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In-House Projects

Glick Automation

<http://34.207.215.191/>

Automation

The screenshot displays a web application for a car dealership. On the left is a 'Filter Inventory' sidebar with the following sections:

- Condition:** A checkbox for 'Used (30)'.
- Make, Model & Trim:** Three dropdown menus for 'All Makes', 'All Models', and 'All Trims'.
- Price:** A range selector with input fields for '\$ 14995' and '\$ 38995', and a red slider bar below.
- Mileage:** A section header with no visible input.

On the right, there are three car listings, each featuring a photo of a car, a 'SALE! \$1,600 DOWN!' banner, and a table of specifications:

Price	Mileage
\$26,995	38,000

Below the table, the specifications are listed:

- Engine: N/A
- Transmission: 6-Speed Shifttable Automatic
- Drivetrain: AWD
- Ext. Color: Green

The other two listings follow a similar format for a 2019 Honda Odyssey and a 2021 Honda Civic Hatchback. The Civic listing has a red circular icon with a white 'D' in the bottom right corner.

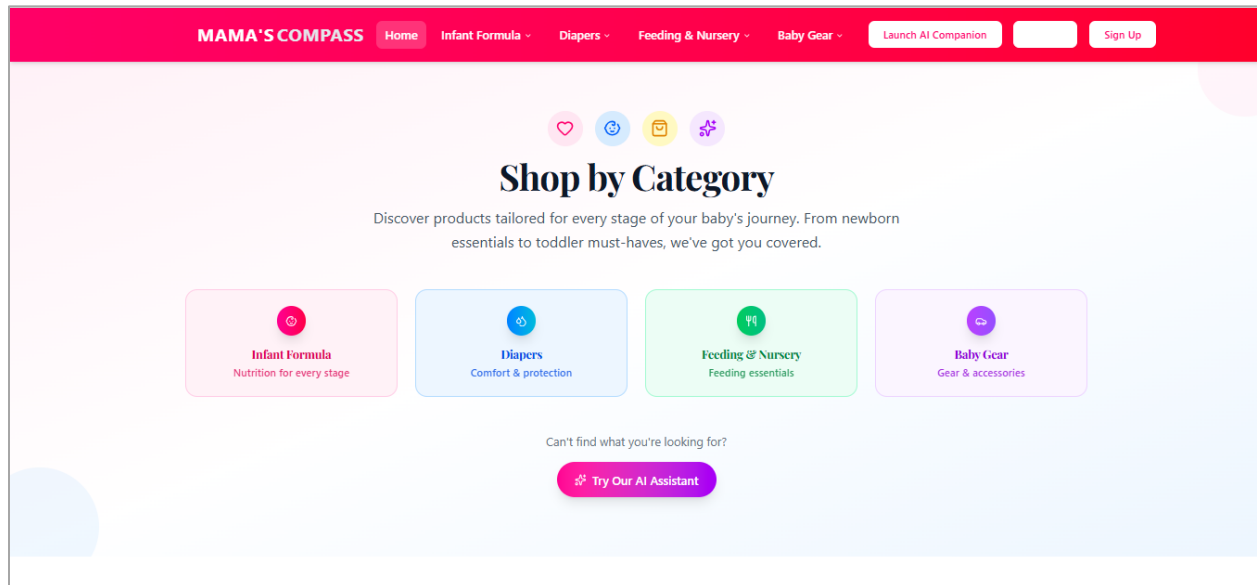
- **Unified customer communications and lead-capture platform**
- Connects **Instagram, Facebook, WhatsApp, and embedded web forms** into a single n8n + GPT-5 workflow.
- Enables **natural GPT-5 conversations** that can escalate into **structured application forms**.
- Triggers **instant workflows** to:
 - Store leads
 - Create records
 - Schedule meetings
 - Log chat history in a central CRM
- **Enforces backend data filters** and **role-based access** to ensure users see only permitted records.
- **Sensitive fields remain gated** for security.
- **Centralizes multichannel messaging** and prevents duplicates.
- Maintains **predictable automation, one clear response per request, unique-submission enforcement, and secure session handling**.

Tech Stack: PostgreSQL, Web CRM (custom), Instagram/Facebook/ WhatsApp integrations, Embedded form/chat widget, Session-based auth.

Mama's Compass

<http://3.94.123.48/>

E-Commerce



- **AI-powered e-commerce aggregation platform.**
- **Collects and normalizes product data** from major marketplaces (starting with Amazon).
- Allows shoppers to **compare products, specifications, prices, stock, and reviews** in one place.
- **Generates SEO-friendly blog content** and **product-focused copy** using OpenAI.
- **Attaches affiliate links** to monetize clicks and conversions.
- Provides **contextual/product-aware chat** for users.
- Offers **personalized recommendations** for registered users.
- **Simplifies product discovery** for shoppers.
- **Streamlines affiliate monetization** for marketers.
- **Centralizes data ingest, normalization, content generation, and performance tracking.**

Tech Stack: FastAPI, Next.js, PostgreSQL (RDS), AWS EC2, Docker, Nginx, JWT + RBAC, RapidAPI, OpenAI, CI/CD, Google-quality SEO & monitoring tools.

Settle In

<https://settlein-tawny.vercel.app/onboarding>

Automation



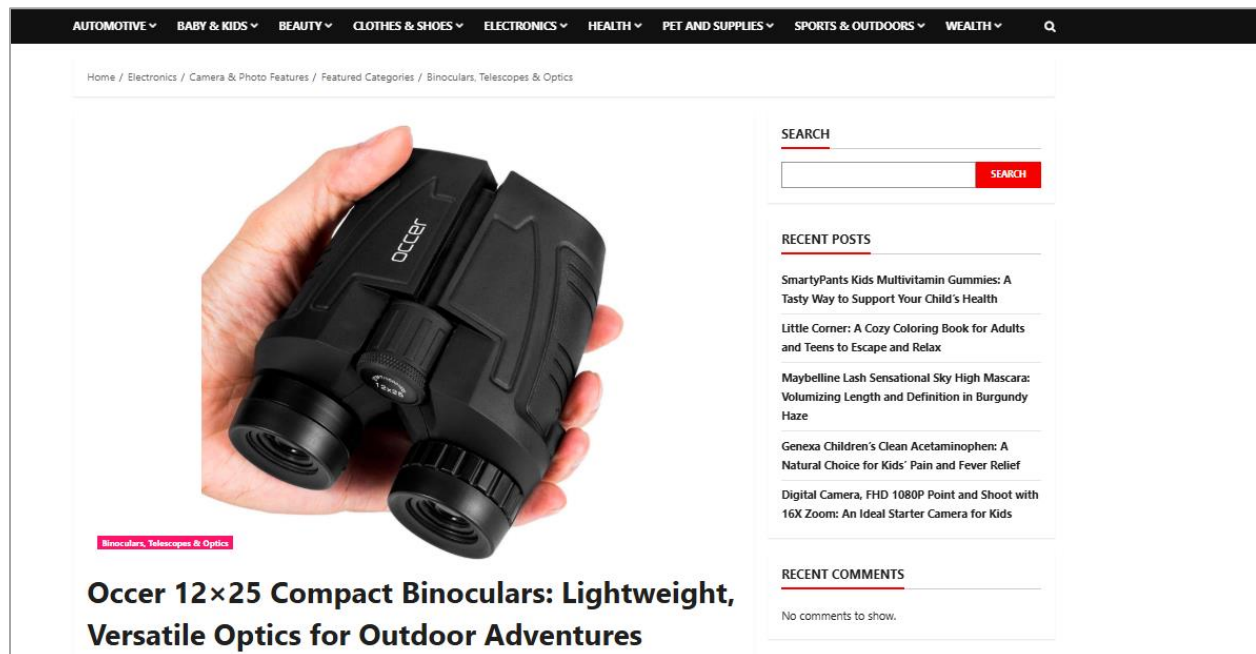
- **Digital advisory platform** for new immigrants in the U.S.
- Helps users **assess eligibility, financial readiness, and documentation requirements**.
- Built with a **modern, mobile-first interface** for accessibility and ease of use.
- Users can:
 - **Chat** with advisors or AI assistants.
 - **Upload ID and bank documents** directly through the platform.
 - Receive **instant, rules-based scores** derived from predefined **Excel criteria**.
- Utilizes **AWS serverless OCR** for **automatic data extraction** from uploaded documents.
- Integrates **AI-assisted scoring** to provide **instant, data-driven feedback**.
- Features a **secure admin dashboard** for human review and submission management.
- Provides **full visibility** to both users and admins into submissions, extracted data, and scoring results.
- **Simplifies the immigration onboarding process** by automating manual steps.
- Ensures **data accuracy, transparency, and compliance** while reducing advisory workload.

Tech Stack: React, Node.js, AWS Serverless, Vercel, OCR, OAuth (Google & Microsoft), MySQL

DropPR Intelligent Product Comparison Platform

<https://codesquada.co/>

E-Commerce



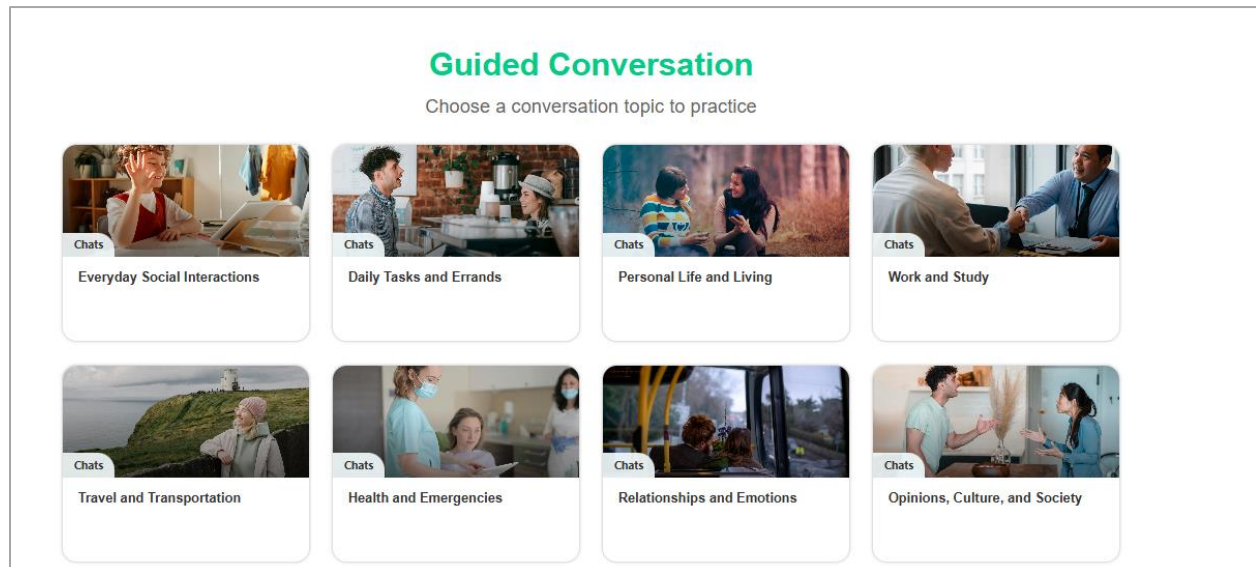
- **AI-powered comparison platform** for finding identical products across major retailers. Amazon, BestBuy, Target, and Walmart.
- Uses **UPC/EAN barcodes** as the **canonical identifier** for accurate product matching.
- **Normalizes product attributes** to ensure consistent cross-platform comparisons.
- Compares **real-time prices, stock availability, specifications, and aggregated reviews**.
- Displays **side-by-side product comparisons** for clear decision-making.
- Integrated **GPT-powered content engine** generates:
 - **SEO-optimized product reviews**
 - **Comparison articles**
 - **Buying guides**
- Includes **embedded affiliate links** for seamless **monetization and tracking**.
- **Designed to eliminate confusion** from cross-site shopping and help users **choose the best deal confidently**.
- Supports **affiliate-based revenue growth** through **automated AI content generation and analytics**.

Tech Stack: React (frontend), backend APIs (retailer & affiliate integrations), real-time price & stock schedulers, data normalization + caching, GPT-based content generation (OpenAI), analytics & admin dashboard.

Gaeilgeoir AI-Powered Irish Language Learning Platform

<https://learn.gaeilgeoir.ai/conversation>

Learning



- **AI-driven language learning platform** focused on helping users master the **Irish (Gaeilge) language**.
- Uses **gamified lessons**, **real-life conversations**, and **adaptive feedback** to enhance engagement and fluency.
- Prioritizes **cultural understanding and conversational fluency** over rote memorization.
- Learners participate in **challenges, quizzes**, and **voice-based dialogue practice**.
- Includes **AI-powered pronunciation feedback** and **dynamic scenario generation** for realistic speaking experiences.
- Offers a **personal performance dashboard** to:
 - Track learning progress
 - Visualize achievements
 - Provide personalized recommendations
- Features **specialized modules for the Irish Leaving Cert oral exam**, including:
 - Mock tests
 - Examiner-style evaluations
- **Designed to make Irish learning immersive, enjoyable, and measurable**, blending **education, culture, and smart technology**.

Tech Stack: React, Node.js, OpenAI (GPT-based dialogue & feedback), Speech Recognition APIs, PostgreSQL, Stripe (subscriptions), and cloud-based video integration.

RTM Kinetics Remote Therapeutic Monitoring Platform

<https://d26f1dm6kt7m07.cloudfront.net/>

Healthline



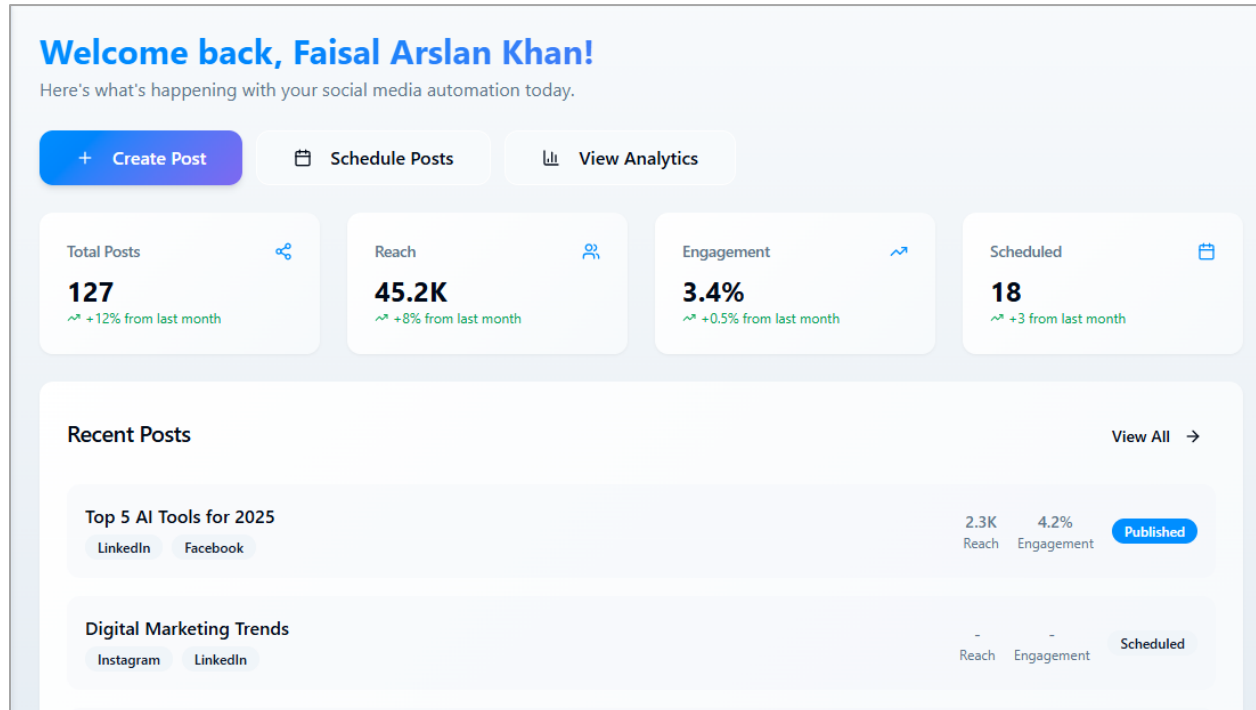
- **Digital platform** connecting **therapists and patients** for remote rehabilitation and exercise management.
- Enables **remote exercise monitoring**, **secure communication**, and **real-time progress tracking**.
- **Patients** can:
 - Access **guided exercise programs** with **video demonstrations**
 - **Log daily activities** and automatically sync data to therapist dashboards
- **Therapists** can:
 - Assign **personalized exercise programs**
 - **Monitor adherence** and performance in real time
 - Communicate via **secure chat or video calls**
 - Manage **billing, scheduling, and reporting** within the same platform
- Includes **live exercise monitoring** to ensure proper technique and prevent injury.
- Captures **detailed performance data and analytics** for tracking recovery trends.
- **Designed to improve patient engagement, reduce therapist workload, and enhance remote therapy effectiveness.**

Tech Stack: React/Node.js, HIPAA-compliant video & messaging, PostgreSQL, real-time monitoring tools, analytics dashboards, payment integration.

Social Media Post Automation PSirondak

<http://18.216.93.155/login>

Automation



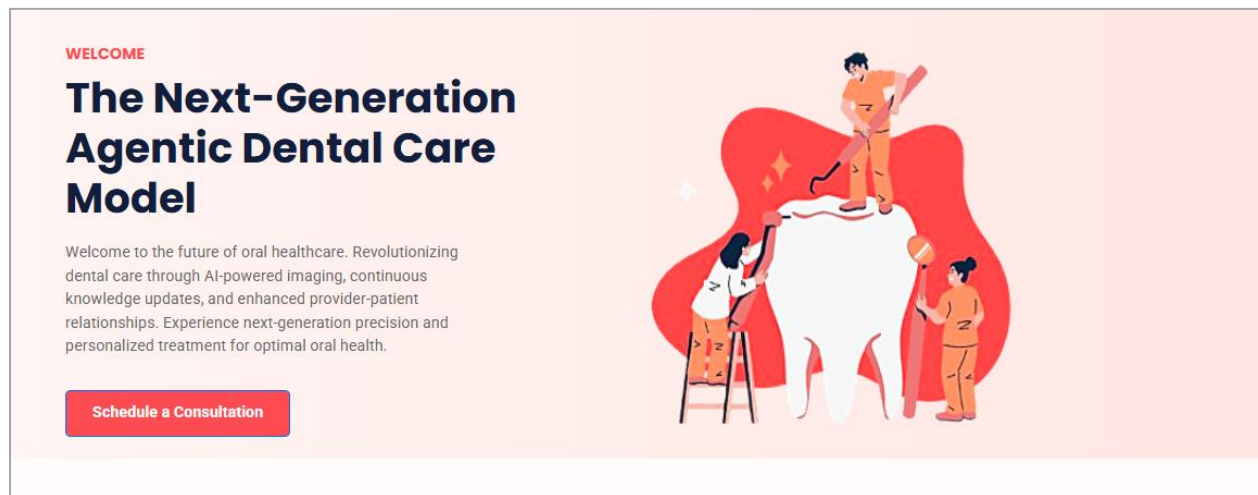
- **AI-powered automation platform** for managing content creation and publishing across **LinkedIn, Facebook, Instagram, and WordPress**.
- Allows users to **generate platform-optimized captions, hashtags, and images** using **OpenAI**.
- Provides a **unified dashboard** for users to **review, edit, approve, and schedule posts** efficiently.
- Ensures **consistent branding** and **streamlined workflows** across multiple platforms.
- Supports **secure account linking and authentication** via OAuth 2.0 and token-based authorization.
- Enables **automated publishing with post tracking** and detailed activity logs.
- Stores media files securely using **cloud storage (AWS S3)**.
- Built on a **scalable backend** with **FastAPI** and **PostgreSQL** for high performance and reliability.
- **Designed to reduce manual posting**, save time, and maintain brand consistency across all social channels.

Tech Stack: Fast API, PostgreSQL, SQL Alchemy, OpenAI (text & image generation), AWS EC2 & S3, OAuth 2.0, JWT authentication, multi-platform API integrations, scheduling & logging infrastructure.

Frontline Dental AI-Powered Dental Care Platform

<https://codesquad-ai.github.io/Frontline-Dental/>

Healthline



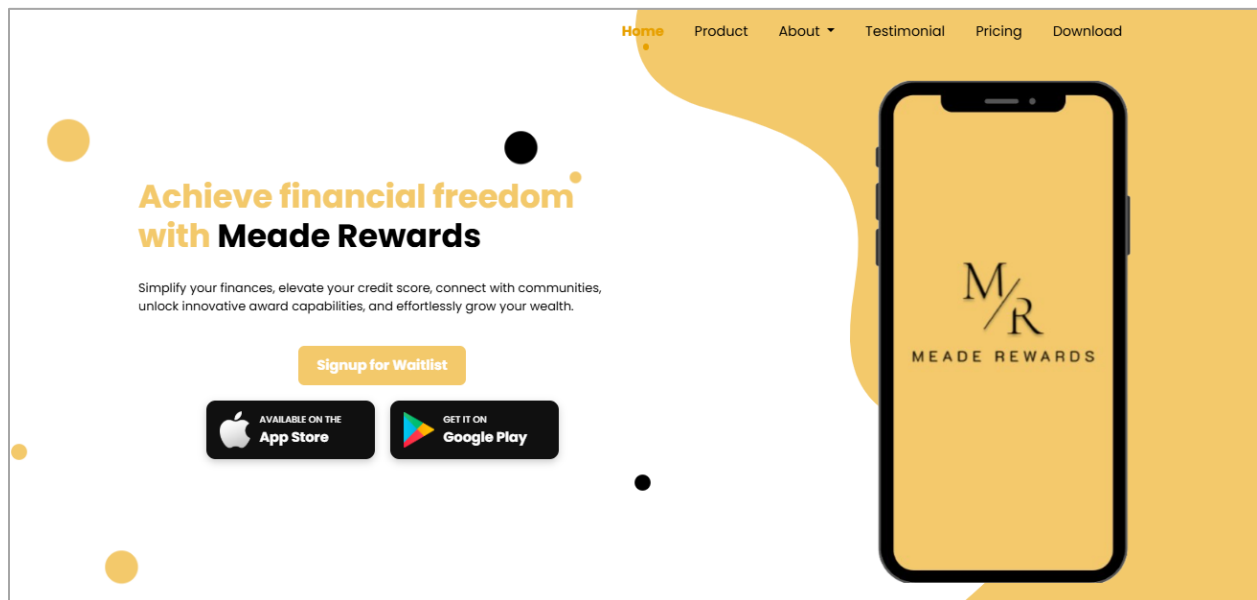
- **Unified digital platform** combining **diagnostics, treatment planning, telehealth, and clinic management** in one intelligent ecosystem.
- **AI-enhanced imaging analysis** detects **caries, fractures, and gum irregularities** from dental scans.
- **Predictive treatment engine** recommends **personalized care pathways** based on patient history and risk profiles.
- **Patient portal** enables **secure virtual consultations**, follow-ups, and real-time communication.
- **Risk-based preventive care system** scores patients dynamically and automates preventive alerts.
- **Automated inventory and insurance management** streamlines stock tracking and claim submissions.
- Implements **end-to-end data security** with **HIPAA/GDPR compliance** and multi-level user access control.
- **Designed to reduce manual errors**, improve operational efficiency, and deliver **data-driven, personalized dental care**.

Tech Stack: AI/ML for imaging analysis, predictive modeling, React/Node.js for portal/dashboard, secure telehealth integration, PostgreSQL, HIPAA/GDPR-compliant encryption, automated inventory and billing modules.

Meade Rewards Intelligent Financial Empowerment Platform

<https://codesquad-ai.github.io/MEADE-REWARDS/>

Finance



- **AI-powered fintech platform** that unifies **financial management, credit insights, investment planning, and business growth tools**.
- Provides **personalized recommendations** for budgeting, spending, and saving through intelligent analytics.
- Offers **credit score tracking and improvement tools** with real-time progress insights.
- Includes an **investment planning dashboard** featuring **AI-driven risk assessment** and portfolio optimization.
- Enables users to **discover funding programs, grants, and loans** based on eligibility profiles.
- Features a **membership and mentorship system** connecting users with certified financial advisors.
- Integrates a **24/7 AI chatbot** for financial guidance, investment advice, and platform assistance.

- Showcases a **modern, conversion-optimized landing page** highlighting platform benefits and membership plans.
- **Designed to simplify financial management**, build user confidence, and promote **long-term financial growth**.

Tech Stack: AI/ML analytics, NLP-based chatbot, React/Node.js frontend and backend, PostgreSQL, secure financial data APIs, interactive dashboards, SEO-optimized landing page, payment & membership management modules.

Cold Email Outreach

[Watch Demo](#)

Automation

- **Automated lead generation and email outreach system.**
- Converts a simple **form input** (category, city, country, campaign) into verified business leads.
- **Scrapes Google Maps** to find businesses and collect their websites.
- **Extracts and verifies emails** automatically.
- Saves results to **Google Sheets**.
- Pushes verified leads to **Instantly** for multi-account email campaigns with **scheduled follow-ups**.
- **Eliminates manual prospecting** and improves data accuracy.
- Runs **end-to-end outreach at scale**.
- Handles **duplicate prevention** and **verification-cost optimizations**.
- **Key reliability fixes:**
 - Switched to **Hunter.io** for faster extraction/verification.
 - Limits **re-verification** to new emails only.
 - Filters **duplicates before sending** to Instantly.

Tech Stack: n8n, Apify Google Maps Scraper API, Hunter.io, Instantly, Google Sheets, Gmail/SMTP, optional tools tested (Jina.ai, other scrapers).

Job Nimbus-Phase 1

[Watch Demo](#)

Automation

- Fully automated workflow system capturing user data via **Telegram**.
- Converts Telegram messages into **structured forms in n8n**.
- **Verifies and updates records** automatically.
- Automates **end-to-end job management in Job Nimbus**, including:
 - Job creation
 - Contact creation
 - Calendar scheduling
 - Google Drive folder organization
 - Presentation updates
 - Email notifications
- **Monitors Google Drive folders** to automatically insert new images into presentations.
- Handles **meeting reschedules** by updating calendars and notifying participants.
- Uses **OpenAI and Cloud-Mersive** for intelligent validation and media conversion.
- Incorporates a **human-correction loop** for data quality in low-confidence steps.
- **Eliminates manual data entry**, reduces errors, and keeps jobs, communications, and client assets **fully synchronized**.

Tech Stack: Telegram Bot, n8n, Job Nimbus API, Gmail API, Google Calendar API, Google Drive API, Presentation API, Cloud-Mersive, OpenAI.

Job Nimbus-Phase 2

[Watch Demo](#)

[Automation](#)

- Automates **internal intake and mission forms** (TP Intake & HO Mission).
- Built using **JotForm + n8n** automation workflows.
- Automatically creates **Job Nimbus jobs** for TP Intake submissions.
- Populates **structured Google Sheets**, including:
 - Main **Template sheet**.
 - **Sub-sheets** like “Address and Names.”
- Sends **real-time Telegram notifications** upon completion.
- **HO Mission** flow:
 - Verifies contact existence before processing.
 - Creates sub-sheet templates only when contact exists.
- **Prevents duplicate records** through contact validation logic.

- Includes **Telegram confirmations** for successful completion of workflows.
- Streamlines **internal workflows** and reduces manual data entry.
- Enhances **data integrity** and **communication speed** across teams.

Tech Stack: JotForm, n8n, Job Nimbus API, Google Sheets, Telegram Bot, Google Drive.